

DIRECT VERTICAL CONNECTION INSTALLATION ANALYSIS PROCEDURE

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1. Purpose & Scope**2. References****3. Definitions****4. Roles & Responsibilities****5. Procedure****6. Records****7. Appendices & Forms**

DETAILED DESCRIPTION OF CHANGES

Revision	Detailed Description of Changes	Change Action Code*	MOC Required?*
0	First issue	N/A	No

***LEGEND:**

CHANGE ACTION CODE	DESCRIPTION
N/A	New Release
A	Typographical / grammatical correction; formatting change; text clarification; no action required
B	Change requiring minor adjustment to process and/or records; users to be informed; no retraining required
C	Major change to process and/or records; retraining of involved personnel to be conducted

MOC REQUIRED?	DESCRIPTION
YES	Change has potential risk related to project, product quality, delivery and service execution. <i>Note: This document is not an MOC Form. An official initiation of MOC should be completed in the applicable system/tool.</i>
NO	Change does not have potential risk related to project, product quality, delivery and service execution.

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*(For optional sections, keep the section present and type “Not Applicable” if the section is not used.
This will allow for potential future use.)*

1. PURPOSE & SCOPE

This procedure applies to the Installation Analysis Subsea (IAS) department to be a reference and a guide to Direct Vertical Connection (DVC) global analysis. The document aims to present a complete methodology for 1st and 2nd ends DVC analysis covering all different kinds of contracts: LTC, EPCI and Supply projects. Each contract has a different approach, however all of them will be gathered in this procedure.

2. REFERENCES

Reference		Description
[1]	ET-3010.00-1500-941-PLR-004 Rev.0	Análise de cargas de dutos em equipamentos submarinos
[2]	ET-3000.00-1500-941-PMU-006-Rev.C	Metodologia e Diretrizes para Análise de Carga em MCV
[3]	ET-3010.00-1500-941-PZ9-001 Rev.D	PROCEDIMENTO PARA ANÁLISE DE VIABILIDADE DE CONEXÃO VERTICAL SEM RESTRITOR DE CURVATURA
[4]	OrcaFlex Help	OrcaFlex Help

3. DEFINITIONS

Acronyms	Description
A&R	Abandonment and Recovery
DVC	Direct Vertical Connection
EPCI	Engineering, Procurement, Construction, and Installation
IAS	Installation Analysis Subsea
LTC	Long Term Charter
MBR	Minimum Bend Radius
PLET	Pipeline End Termination
PLEM	Pipeline End Manifold
PLSV	Pipe Laying Support Vessel
UTM	Umbilical Termination Module
VCM	Vertical Connection Module

Abbreviations	Description
H_s	Significant Wave Height
H_{max}	Maximum Wave Height
s	Seconds

4. ROLES AND RESPONSIBILITIES

Responsibility and activity are summarised in the following matrix:

Responsibility Matrix Legend: R = Responsible for performing the action A = Accountable to ensure action happens. Only one A in case of more R C = Consulted during the action I = Informed after the action has completed	Requester	Installation Analysis Executor	Installation Analysis Reviewer	Installation Analysis Lead	Installation Analysis Principal
1. Open job requisition and supply input data	R	I	I	I	-
2. Consolidate information and fill checklist to ensure analysis can be carried out	C	R / A	-	A	-
3. Modelling on finite element interface	-	R / A	-	A	C
4. Perform analysis, fill analysis checklist and send to internal review	-	R / A	-	A	C
5. Review analysis model and methodology and fill checklist	-	I	R	R / A	C
6. Prepare analysis report	-	R	-	I	-
7. Internal analysis report review	-	I	R	-	C
8. Interdisciplinary Check (IDC)	R	I	I	I	-
9. Report approval	I	I	I	R / A	R
10. Report issue	I	R	R	R / A	-

Notes:

Note 1: Installation Analysis Executor could be any installation analysis engineer or assistant. An intern could also be executor under supervision.

Note 2: Installation Analysis Reviewer must have a higher level of proficiency and expertise compared to the executor. In the event of high workload or team vacations, the role may also be assumed by the Installation Analysis Lead.

Note 3: Installation Analysis Lead responsibilities may be assumed by installation analysis manager or principal in their absence.

5. PROCEDURE

5.1 INTRODUCTION

Direct Vertical Connection (DVC) operation enables flexible line connection to subsea equipment such as PLEM, PLET and Christmas Trees in deep water environment without the aid of diving operations and / or support vessel. The connection is made by an equipment called Vertical Connection Module (VCM) which is connected to the flexible pipe through a flanged connection. Other equipments can be considered for a DVC operation, such as UTM's for umbilicals. In this document the DVC equipment will be always referred as a typical VCM, as shown in Figure 1.

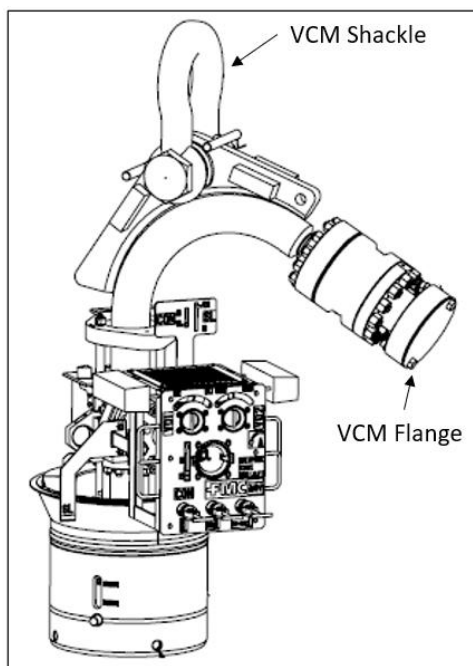


Figure 1: Vertical Connection Module (VCM)

VCM must be verticalized and then connected to subsea equipment. During operation, loads at VCM flange may exceed the design limits due to flexible dynamics. Temporary buoyancy modules or dead weight may be required to be installed along the line to aid the VCM verticalization respecting the product and its ancillaries acceptable limits. Thus, it is required to study the best configuration through finite element method global analysis.

There are two DVC operation possibilities:

- 1st end DVC: VCM is connected to the line 1st end at PLSV working table, and during VCM overboarding the catenary will be supported by PLSV tensioner. At a safe distance above the subsea equipment, PLSV crane is connected to VCM shackle to verticalize the VCM. After verticalization, PLSV approaches the VCM towards the hub and performs the connection, as shown in Figure 2.

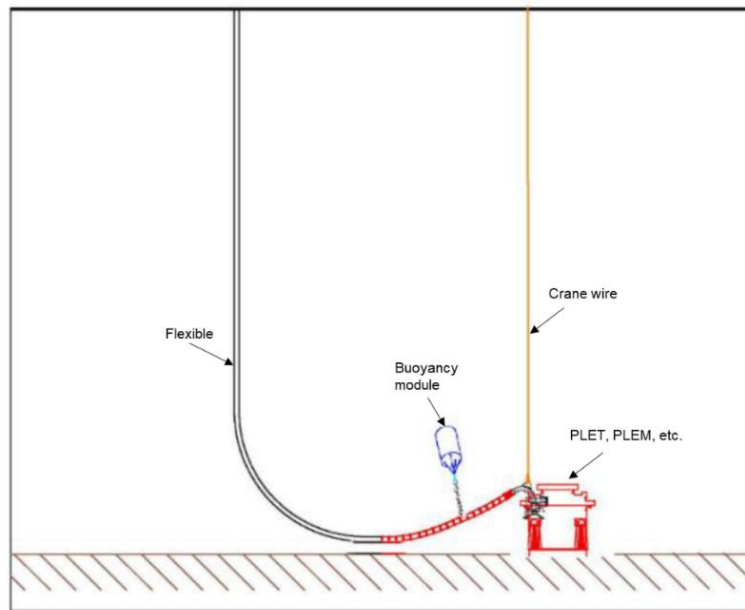


Figure 2: 1st End DVC

- 2nd end DVC: Line 2nd is connected to the VCM at PLSV working table and during VCM overboarding, the catenary will be supported by A&R wire. When VCM is close to seabed, ROV connects the PLSV crane wire to VCM shackle, and load is transferred from A&R to PLSV crane. Then ROV connects A&R wire to the pre-assembled slings on the line for the camelback formation as shown in Figure 3.

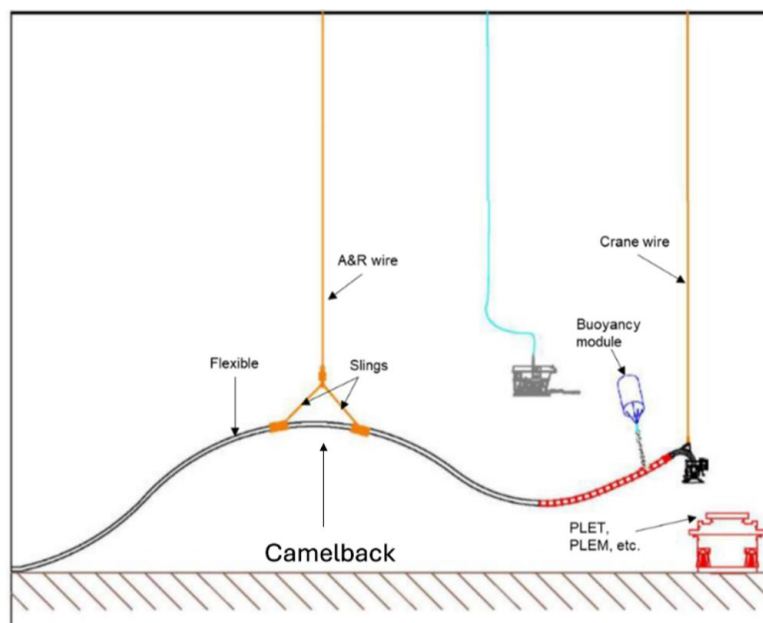


Figure 3: 2nd End DVC

5.2 REQUIRED DATA FOR CARRYING OUT THE ANALYSIS

Prior to start of the analysis, it is necessary to have access to the following project data:

- General:
 - PLSV to perform the DVC
 - Water Depth
- Flexible Line:
 - External Diameter
 - Internal volume
 - Mass with empty and full bore, in air and in water
 - Annulus volume
 - Axial stiffness
 - Torsional stiffness
 - Bending stiffness at 20°C (for umbilicals only)
 - Hysteretic bending stiffness curve – Curvature x Bend moment (applicable with the evaluated installation conditions)
 - Seabed friction coefficients: Lateral and Axial
 - Bore condition: full or empty
 - Annulus condition: intact or damaged external sheath
 - Minimum allowable bend radius during installation (see notes)

Notes:

Note 1: The MBR value must be verified with the project. In the absence of any exceptions (for instance: gas export or different materials pipes), the default value to be considered is:

$$MBR_{installation} = 1,25 * MBR_{storage}$$

Note 2: According to Ref.[3], for DVC analysis with no bend restrictor (or with covered angle lower than 72°), the flexible MBR must be considered as:

$$MBR_{installation} = 1,5 * 1,25 * MBR_{storage} = 1,875 * MBR_{storage}$$

- End Fitting:
 - Total length
 - Contact diameter
 - Internal diameter (it must be modeled as the same flexible line internal diameter)
 - Mass in air and apparent submerged weight
- Bend Restrictor:
 - Total length
 - Contact diameter
 - Mass in air and apparent submerged weight
 - Locking radius

- If applicable, bend restrictor capacity curve (bend moment x shear force) should be considered
- Rings Length (see note 1)
- Bend restrictor bend stiffness curve (see note 2)

Notes:

Note 1: Bend restrictor in OrcaFlex shall be modelled with a number of finite elements equal to the number of the bend restrictor's rings.

*Note 2: If not sent by the project team, the Bend stiffness of the bend restrictor is estimated directly by its geometry. Please check in **Appendix 2 – Bend restrictor bending stiffness calculation**.*

- Vertical Connection Module (VCM)

- Dimensions (see note)
- Gooseneck angle
- Weight in air and apparent submerged weight

Note:

*Note 1: **Appendix 1 – VCM data** - shows all required dimensions for the VCM modelling.*

- Buoyancy Modules

- Mass and nominal buoyancy
- Drag area, drag coefficient, and added mass coefficient (see note)

Notes:

Note 1: The exact hydrodynamic coefficients are just required to be modelled if supply analysis is performed. The properties of the buoys to be considered are suggested in section 5.3 from ref. [1] as shown in Table 1. The supply of this buoyancy modules is not within the TechnipFMC scope (supplying flexible lines).

Table 1 – Supply analysis DVC buoyancy modules properties

Nominal Buoyancy (tf)	Mass (t)	Volume (m³)	Compressibility Modulus (kPa)	Height (m)	Drag Area (m²)			Drag Coefficient (Cd)			Addes Mass Coefficient (Ca)		
					X	Y	Z	X	Y	Z	X	Y	Z
0,1	0,1429	0,2369	Infinity	0,8788	0,5149	0,5149	0,2696	1,0	1,0	0,88	0,657	0,657	0,637
0,2	0,2857	0,4739	Infinity	1,1073	0,8174	0,8174	0,4280	1,0	1,0	0,88	0,657	0,657	0,637
0,2	0,3571	0,5923	Infinity	1,1928	0,9485	0,9485	0,4966	1,0	1,0	0,88	0,657	0,657	0,637
0,3	0,4286	0,7108	Infinity	1,2675	1,0710	1,0710	0,5608	1,0	1,0	0,88	0,657	0,657	0,637
0,4	0,5714	0,9477	Infinity	1,3951	1,2975	1,2975	0,6794	1,0	1,0	0,88	0,657	0,657	0,637
0,5	0,7143	1,1847	Infinity	1,5028	1,5056	1,5056	0,7883	1,0	1,0	0,88	0,657	0,657	0,637
0,75	1,0714	1,7770	Infinity	1,7203	1,9729	1,9729	1,0330	1,0	1,0	0,88	0,657	0,657	0,637
1,0	1,4286	2,3693	Infinity	1,8934	2,3899	2,3899	1,2514	1,0	1,0	0,88	0,657	0,657	0,637
1,25	1,7857	2,9617	Infinity	2,0396	2,7733	2,7733	1,4521	1,0	1,0	0,88	0,657	0,657	0,637
1,5	2,1429	3,5540	Infinity	2,1674	3,1317	3,1317	1,6398	1,0	1,0	0,88	0,657	0,657	0,637
1,75	2,5000	4,1463	Infinity	2,2817	3,4707	3,4707	1,8172	1,0	1,0	0,88	0,657	0,657	0,637
2,0	2,8571	4,7387	Infinity	2,3855	3,7938	3,7938	1,9864	1,0	1,0	0,88	0,657	0,657	0,637

- Other data:
 - Anodes data (total weight and positions)
 - Other ancillaries as spools or flange adaptors (weight, length, and contact diameter).

5.3 DVC ANALYSIS METHODOLOGY

This section presents the DVC analyses methodology. The specific procedure to do each step will be described in section 5.4. The IAS DVC analyses can be split in 2 phases:

- Static and quasi-static analysis.
- Dynamic analysis.

Additionally, contingencies cases are considered to address urgent operational requirements.

5.3.1 STATIC AND QUASI-STATIC ANALYSIS

The static/quasi-static analysis is split in the following load cases, as per Ref. [2]:

- Case 1: During 2nd end DVC overboarding, catenary load supported by VCM flange and A&R winch wire connected to VCM padeye.
- Case 2: Referred as 1st end DVC static configuration (VCM verticalized and ready to perform the connection). Buoyancy modules may be required.
- Case 3i: 1st end DVC quasi-static analysis simulating a heave-up in the PLSV during a wave incidence after the connection.
- Case 3ii: 1st end DVC quasi-static analysis simulating the moment when the flexible touches the seabed, called as lay away.
- Case 4: 1st end DVC hydrostatic test after the flexible line is laid on seabed.
- Case 5: Flexible line already in operation after buoyancy modules removal.

Note:

Note 1: The quasi-static cases 2 to 5 can be done for the 2nd end DVC.

The specific procedure to do each static/quasi-static step will be described in section 5.4.

5.3.2 DYNAMIC ANALYSIS

Dynamic analysis is split in the following load cases:

- 1st end DVC:
 - VCM at 50m from the seabed.
 - VCM verticalization.
 - VCM connected to the subsea equipment.
- 2nd end DVC:
 - End fitting at PLSV worktable.
 - VCM overboarding.
 - Waiting for crane wire connection.
 - VCM connected to subsea equipment.

The specific procedure to do each dynamic step will be described in section 5.4.

5.3.3 CONTINGENCY ANALYSIS

Contingency analyses are carried out to assess the moment where the VCM will be connected, but for any problem, the flexible line must be lifted to ensure a more secure engagement of the VCM with the hub. The contingencies cases are:

- Vertical loads for lifting the flexible line during the soft landing – Applied to 1st and 2nd end DVC.
- Maximum camelback pull – Applied to 2nd end DVC.

The specific procedure to do each contingency case will be described in section 5.4.

5.4 DVC ANALYSIS PROCEDURE

The procedure to perform each analysis case listed in section 5.3 is described in this section. The flange loads reference to be followed is shown in Figure 4.

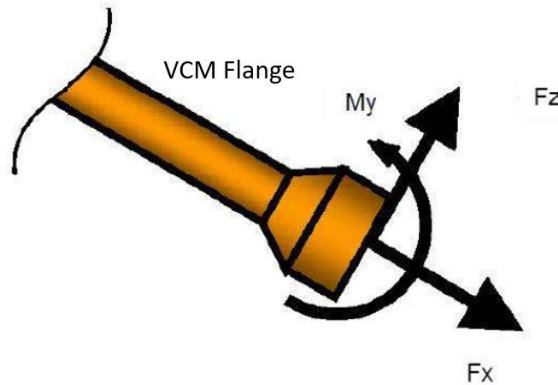


Figure 4: VCM Flange Reference System

5.4.1 MODELLING

DVC analysis shall be performed using the finite element software OrcaFlex. Flexible lines or umbilical and end-fitting shall be modelled as line types. Bend restrictor shall be modelled as attachment type. VCM shall be modelled as 6D buoy. PLSV crane can be modelled as line type or winch. Buoyancy modules can be modelled either as 3D buoy or attachment type. For 2nd end DVC modelling, the camelback slings shall be modelled as links and the A&R wire as a winch or line type.

In addition, some standards are adopted as:

- OrcaFlex global axes must be at XZ plane, as shown in Figure 5.

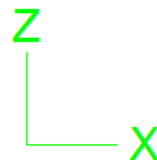


Figure 5: OrcaFlex global Axis to be Considered During Modelling.

- The flexible axis should respect the Orcina recommendations (Ref. [4], section “Line data: End connections”) as shown in Figure 6.

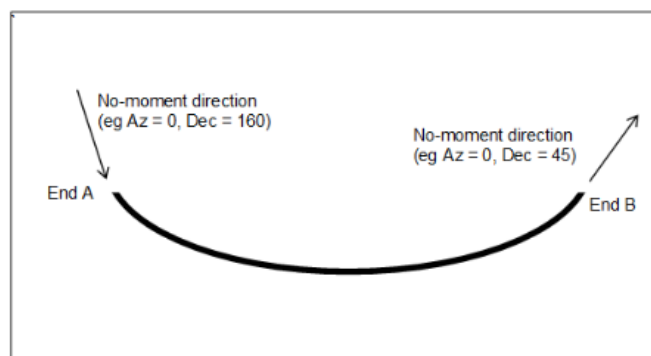


Figure 6: Orcina Flexible Line Axis Orientation

- The Flexible line End A must be connected at the top and the End B at the bottom. Both ends must have the same orientation (Z axis as axial direction and X axis with no torsion) as shown in Figure 7.

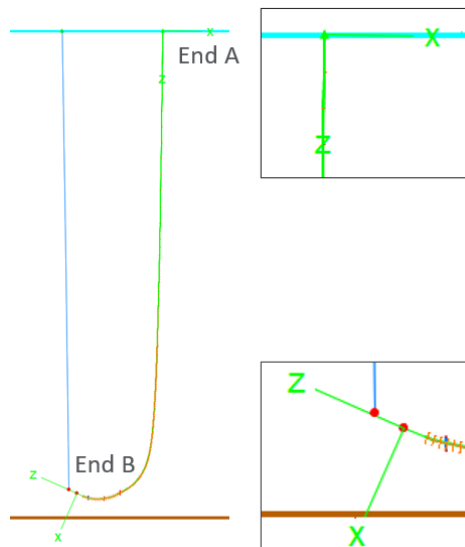


Figure 7: Flexible Line Local Axis Correct Modelling

Notes:

Note 1: Modelling standardization seeks to facilitate results post processing.

Note 2: During flange results post processing (End B), it is required to change the bend moment signal, as Ref.[4], section “Line results: End loads”.

Note 3: For 2nd end DVC modelling, the line connection with the VCM is the End B side in order to standardize with 1st end DVC post processing.

- With exception of umbilicals, flexible lines must be modelled considering the hysteresis behaviour in bending stiffness curves.
- For bend restrictor attachment type data, the connection (“Axial load/Inertia transfer”) must be considered as “over full length” as shown in Figure 8.

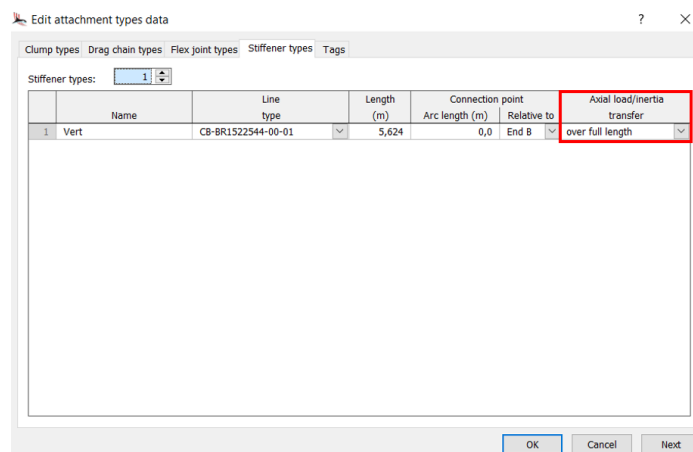


Figure 8: Bend Restrictor connection

- The distance between fireline and crane tip (between flexible line End A and the crane top connection) is usually 25m but can be changed depending on the project.
- The anodes weight shall be considered along the line. It can be modelled as attachments or in bend restrictor and/or end fitting modelling.

- To perform quasi-static analysis, it is not required modelling the PLSV. Instead, to simulate the heave-up movement, a winch object connected to the flexible End A must be modelled with specified length of 10m. To perform dynamic analysis the PLSV shall be modelled considering the conventions used to generate the RAO.

5.4.2 ENVIRONMENT

5.4.2.1 WAVES

The environment conditions to be considered in dynamic analysis must be in accordance with each project. The IAS regular wave approach standard is:

$$H = H_{max} = 1,9 * H_s$$

Where:

H – Regular Wave Analysis Height

H_{max} – Maximum Wave Height

H_s – Significant Wave Height

The regular wave maximum height for DVC analysis must be verified in the project documentation. A standard value to be considered is 3,5m.

The wave periods to be considered as a standard is a range from 6s to 15s with step of 1s but must be checked for each project environment conditions (Metoceans). The standard wave and current incidence directions in relation to the PLSV are shown in Figure 9 (a). For supply analysis, Figure 9 (b) can be considered.

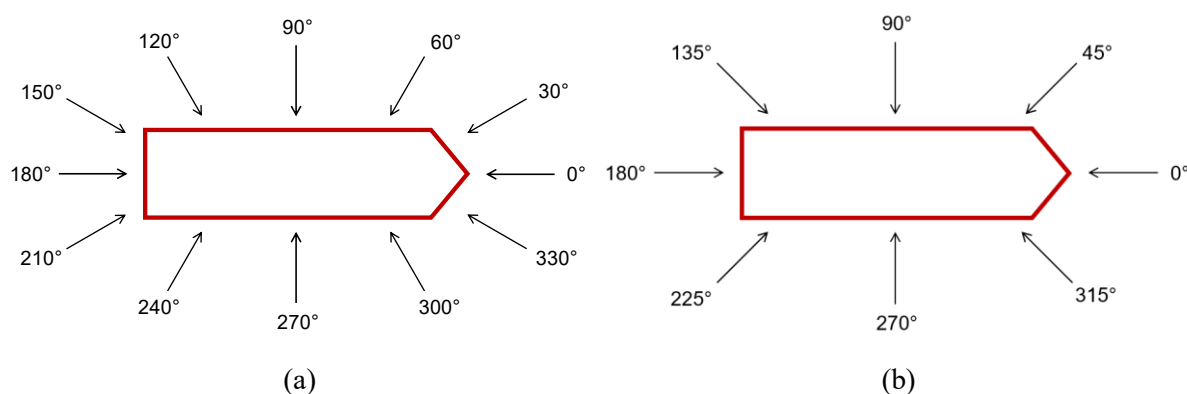


Figure 9: Wave Direction for Dynamic Analysis

5.4.2.2 CURRENT

The specific project 1 year return current profile must be considered. The profile must be selected from the Metocean “Extreme Current Profile” section. The profile with the top high velocity (at water surface or at the minimum water depth presented) must be chosen, as exemplified in Figure 10.

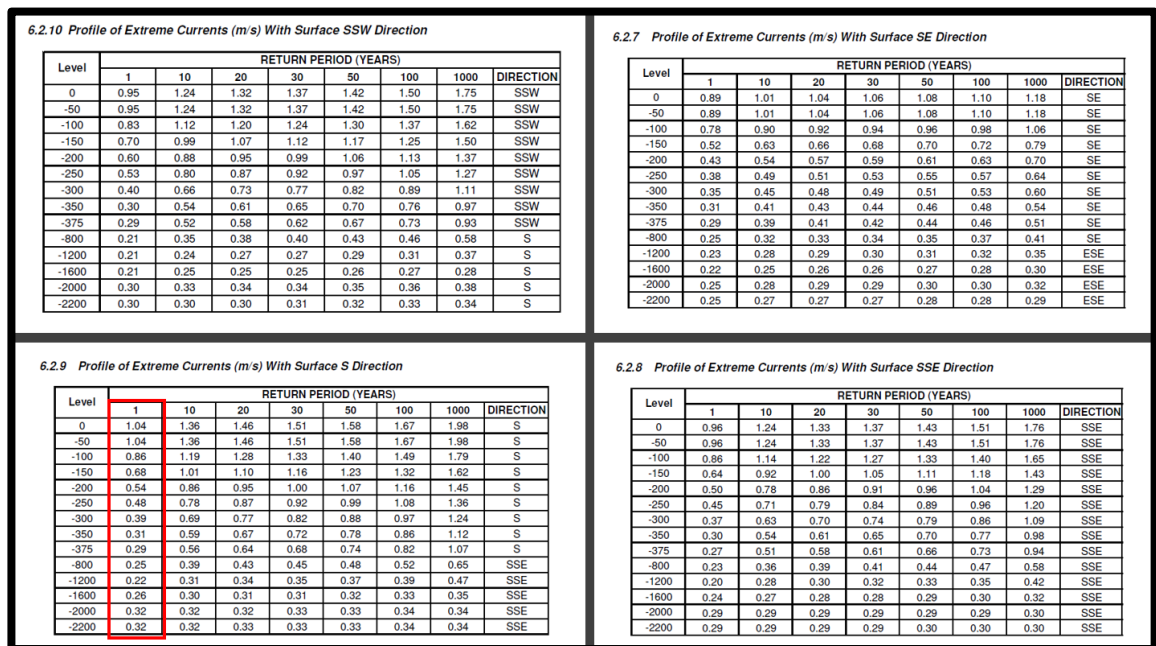


Figure 10: Example of Choosing a Current Profile.

Waves and current must be collinear. Near/Far approach should be used only for operations that the PLSV is free to weathervane, i.e, only for the following DVC steps:

- 2nd end DVC - End Fitting at worktable
- 2nd end DVC - VCM Overboarding

5.4.3 STATIC AND QUASI-STATIC ANALYSIS CASES

5.4.3.1 CASE 1

The loads for the case 1 are determined through a simplified analytical analysis called as ELT (Estimated Laying Tension) by Ref.[1]. No OrcaFlex model is required, and the calculation must be done as follows:

$$ELT = DAF * (A + (WD + 10) * CF * w)$$

Where:

DAF – Dynamic Factor – consider 1,25

A – Weight of VCM and ancillaries

WD – Water Depth

CF – Catenary Factor – consider 1,06

w – Flexible pipe submerged linear weight with full bore

5.4.3.2 CASE 2

In order to define the 1st end DVC static configuration, the engineer must start the analysis trying to verticalize the free VCM with no buoyancy modules or dead weight. If the verticalization is not possible, temporary buoyancy modules or dead weight may be considered installed along the line (as shown in Figure 11) respecting the following criteria:

- Minimum distance between VCM flange and the 1st buoyancy modules of 3m.
- Minimum distance between two buoyancy modules connections of 3m.
- The maximum uplift for buoyancy modules in each point is 2000kg.
- The maximum total dead weight installed along the line is 2000kg.
- Avoid installing buoyancy modules close to the end of the bend restrictor (limit of 0,5m)

Notes:

Note 1: The distances of 3m are an IAS best practice and shall be negotiated with the project if necessary.

Note 2: The number of buoys holding points must be as small as possible to reduce operational time.

Note 3: In case of EPCI / LTC projects it is required to check buoyancy modules availability (specific net buoyancy) with the project team.

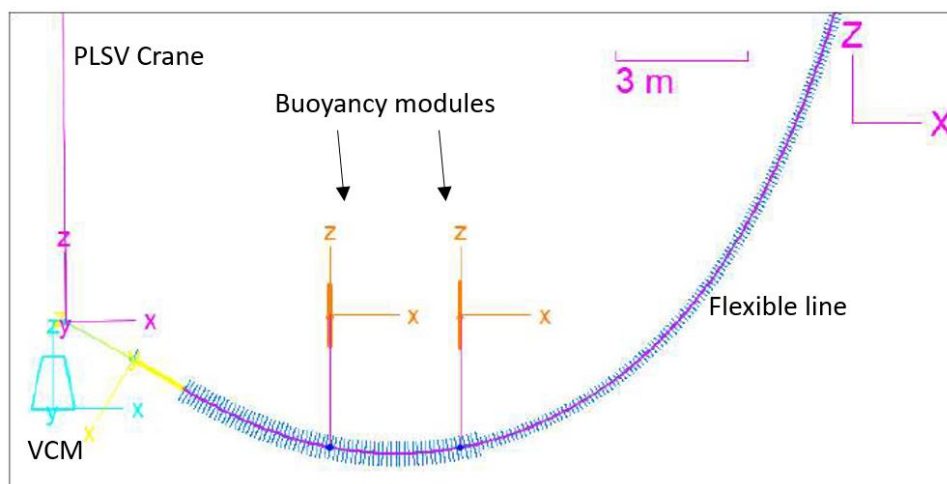


Figure 11: Example of 1st end DVC Static Configuration with 2 Buoys Holding Points.

The static configuration must be found respecting the following limiting criteria:

- Exact VCM Flange height relative to the seabed (presented as “A” in **Appendix 1 – VCM data**).
- Exact horizontal position of VCM Flange if bathymetry is available.
- VCM maximum declination of 0,5° (3° can be adopted depending on the project alignments).
- Flexible minimum seabed clearance of 0,5m.
- Flexible line MBR.
- Bend Restrictor maximum bend moment and shear force.

Notes:

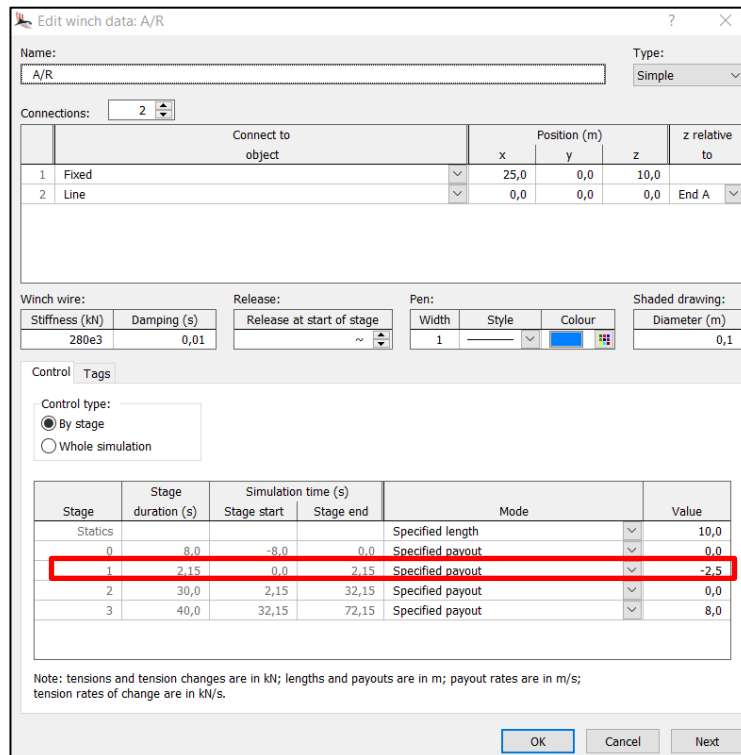
Note 1: The DVC static configuration must be done avoiding the bend restrictor locking, if possible.

Note 2: To align the VCM flange with bathymetry, the X-axis in Seabed origin must be defined with the same value as the VCM Flange X-axis position after the statics simulation.

*Note 3: If it is not possible to respect the 0,5m from line to seabed, dredging may be considered. Dredging quotas can be found in **Appendix 4 – Dredging quotas***

5.4.3.3 CASE 3 (3I AND 3II)

Case 3 must be performed after setting the VCM fixed in the same position of Case 2 configuration. The heave up movement shall be simulated by a winch object connected to the flexible line End A using a specified payout value, as shown in Figure 12. The payout must be as high as possible, starting with 2,5m. It means that the engineer must start the analysis with 2,5m and, if necessary, reduce until the value of 1,8m (as a last attempt, the minimum value of 1,5m can be discussed with your manager). For any heave-up payout value, the entire movement must occur in 2,15s, as Refs.[1] and [2]. This step is called as Case 3i.



Connect to object		Position (m)			z relative to
		x	y	z	
1	Fixed	25,0	0,0	10,0	
2	Line	0,0	0,0	0,0	End A

Stiffness (kN)	Damping (s)	Release at start of stage	Pen: Width	Style	Colour	Shaded drawing: Diameter (m)
280e3	0,01	~	1			0,1

Control type:
☒ By stage
☐ Whole simulation

Stage	Stage duration (s)	Simulation time (s)		Mode	Value
		Stage start	Stage end		
Statics				Specified length	10,0
0	8,0	-8,0	0,0	Specified payout	0,0
1	2,15	0,0	2,15	Specified payout	-2,5
2	30,0	2,15	32,15	Specified payout	0,0
3	40,0	32,15	72,15	Specified payout	8,0

Note: tensions and tension changes are in kN; lengths and payouts are in m; payout rates are in m/s; tension rates of change are in kN/s.

Figure 12: Winch Object Modelled to Apply the Heave-Up Movement.

After heave-up, the simulation must stabilize for a period around 30s (other values can be considered) and then, the lay away should be started paying the winch until the flexible line touches the seabed. This step is called as Case 3ii. The flexible line MBR and bend restrictor loads must be within the limits during whole quasi static period. If the limiting criteria is not respected, a new static configuration shall be found, or the heave-up value (winch specified payout) reduced. Cases 2 and 3 have an iterative sequence, where the following flowchart must be considered:

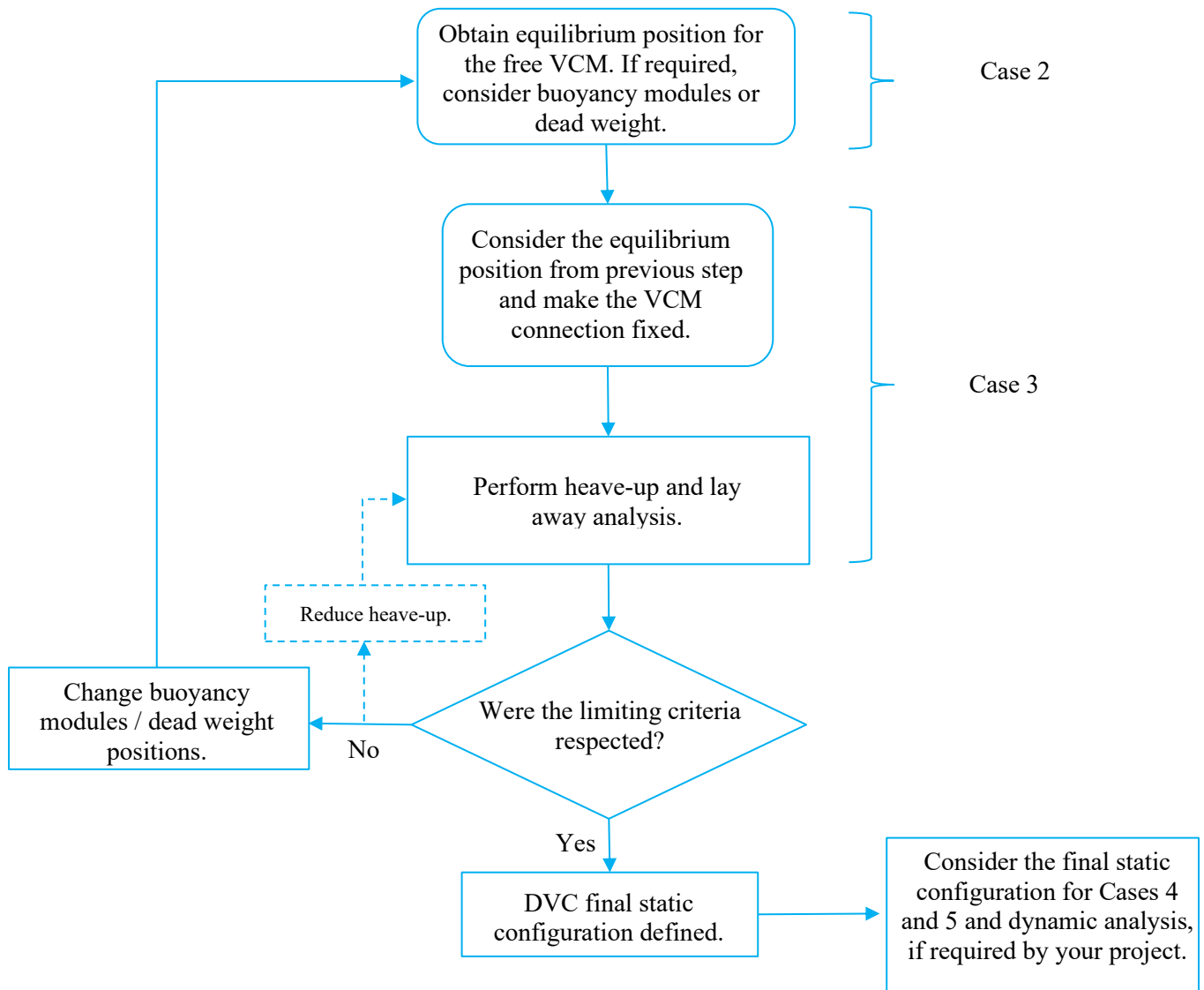
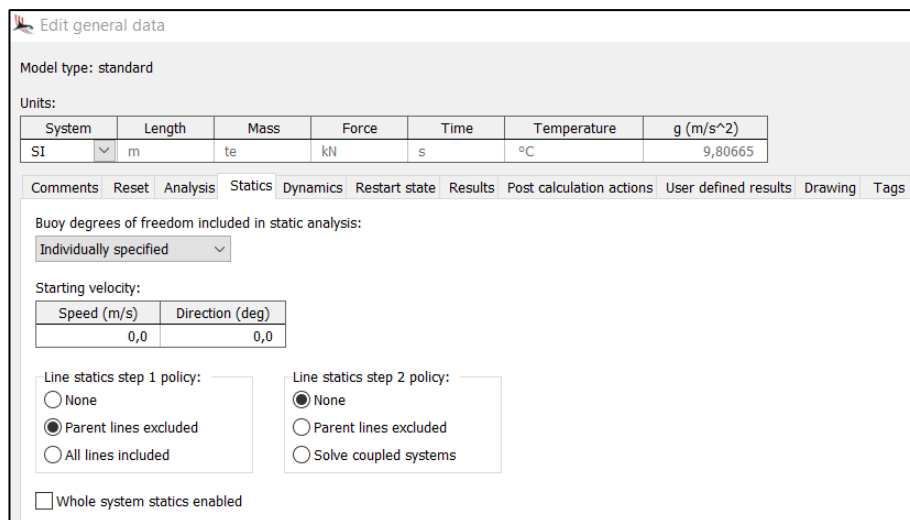


Figure 13: DVC Configuration Flowchart.

5.4.3.4 CASE 4

To start the case 4, the flexible must be around 50m length laid on seabed. Starting from this configuration, follow the steps:

- Save the flexible line nodes positions using “Specified starting shape for lines”.
- Set the winch object (Figure 12) Specified length to its final length (after laying the 50m length).
- Uncheck the box “Whole system statics enabled” in General – Statics, as shown in Figure 14.



Edit general data

Model type: standard

Units:

System	Length	Mass	Force	Time	Temperature	g (m/s^2)
SI	m	te	kN	s	°C	9,80665

Comments Reset Analysis Statics Dynamics Restart state Results Post calculation actions User defined results Drawing Tags

Buoy degrees of freedom included in static analysis:
 Individually specified

Starting velocity:

Speed (m/s)	Direction (deg)
0,0	0,0

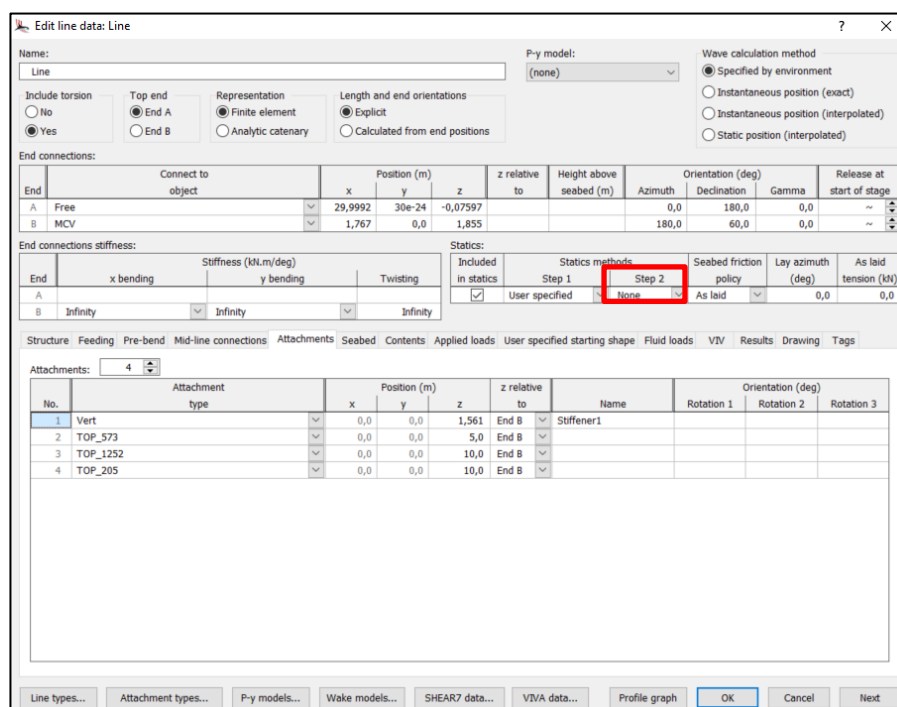
Line statics step 1 policy:
☐ None
☒ Parent lines excluded
☐ All lines included

Line statics step 2 policy:
☒ None
☐ Parent lines excluded
☐ Solve coupled systems

☐ Whole system statics enabled

Figure 14: OrcaFlex Static Configuration to Perform Case 4

- Set the Flexible line static method step 2 as “None”, as shown in Figure 15.



Edit line data: Line

Name: Line

P-y model: (none)

Wave calculation method:
☒ Specified by environment
☐ Instantaneous position (exact)
☐ Instantaneous position (interpolated)
☐ Static position (interpolated)

Include torsion:
☐ No
☒ Yes

Top end:
☒ End A
☐ End B

Representation:
☒ Finite element
☐ Analytic catenary

Length and end orientations:
☒ Explicit
☐ Calculated from end positions

End connections:

End	Connect to object	x	y	z	z relative to	Height above seabed (m)	Orientation (deg) Azimuth	Declination	Gamma	Release at start of stage
A	Free	29,9992	30e-24	-0,07597			0,0	180,0	0,0	
B	MCV	1,767	0,0	1,855			180,0	60,0	0,0	

End connections stiffness:

End	x bending	Stiffness (kN.m/deg)	y bending	Twisting
A				
B	Infinity		Infinity	Infinity

Statics:

Included in statics	Step 1	Step 2	Seabed friction policy	Lay azimuth (deg)	As laid tension (kN)
☒	User specified	None	As laid	0,0	0,0

Attachments: 4

No.	Attachment type	x	y	z	z relative to	Name	Rotation 1	Rotation 2	Rotation 3
1	Vert	0,0	0,0	1,561	End B	Stiffener1			
2	TOP_573	0,0	0,0	5,0	End B				
3	TOP_1252	0,0	0,0	10,0	End B				
4	TOP_205	0,0	0,0	10,0	End B				

Line types... Attachment types... P-y models... Wake models... SHEAR7 data... VIVA data... Profile graph OK Cancel Next

Figure 15: OrcaFlex Static Method to Perform Case 4

- Update the flexible line bending stiffness curve. In this moment, 3 steps will be considered in order to calculate the final Case 4 loads:
 - Step 0 – The bending stiffness to be applied on the flexible line is the same curve from the case 3. Just a static analysis is performed.
 - Step 1 – The bending stiffness to be applied on the flexible line is the same curve from the case 3 but without hysteresis, as shown in Figure 16. A dynamic analysis, with no waves and no current must be run with duration of 50s.
 - Step 2 – The bending stiffness to be applied on the flexible line is the hydrostatic test curve without hysteresis. A dynamic analysis, with no waves and no current must be run with duration of 50s.

Data for FL-101.51796_Flooded

Interpolation method: Linear
 Values outside the table: Extrapolate

Number of rows:

Curvature (rad/m)	Bend moment (kN.m)
0,0	0,0
0,003	0,13
0,0185	0,4087
0,0577	1,0475
0,708	11,6374

☐ Hysteretic

Figure 16: Bending Stiffness Curve Example without Hysteresis

The Case 4 axial load (referred as T), shear force (referred as V) and bend moment (referred as M) are calculated as Ref. [2]:

$$\begin{aligned}
 T &= T_2 - T_1 + T_0 \\
 V &= V_2 - V_1 + V_0 \\
 M &= M_2 - M_1 + M_0
 \end{aligned}
 \quad \text{Equation 1}$$

Where:

T_0, T_1 and T_2 – Axial force for steps 0, 1 and 2 respectively

V_0, V_1 and V_2 – Shear force for steps 0, 1 and 2 respectively

M_0, M_1 and M_2 – Bend moment for steps 0, 1 and 2 respectively

The Case 4 is the moment when the buoyancy modules are removed from the flexible, therefore both conditions must be considered for all cases: with buoys and without buoys.

5.4.3.5 CASE 5

The Case 5 is done in the same way of Case 4 but instead of hydrostatic test bending stiffness curve, it is done with the operations bending stiffness curve, so the steps are:

- Step 0 – The bending stiffness to be applied on the flexible line is the same curve from the case 3. Just a static analysis is performed.
- Step 1 – The bending stiffness to be applied on the flexible line is the same curve from the case 3 but without hysteresis, as shown in Figure 16. A dynamic analysis, with no waves and no current must be run with duration of 50s.
- Step 2 – The bending stiffness to be applied on the flexible line is the operation curve without hysteresis. A dynamic analysis, with no waves and no current must be run with duration of 50s.

The calculation shown in Equation 1 must be used the same way. Case 5 must be done just without buoyancy modules once the operation moment is after removing the buoys.

Notes:

Note 1: If the project does not provide the flexible line contents density, water must be considered for the Case 5. Since the contents density is known, the following guideline shall be considered:

- *Contents density < 1025 kg/m³: consider the flexible line filled with sea water*
- *Contents density > 1025 kg/m³: consider the flexible line filled with real content*

5.4.3.6 2ND END DVC CASES

If required by the project, the 2nd end DVC cases 2, 3, 4 and 5 can be performed. The following particularities must be followed:

- The heave-up is applied by the A&R winch wire connected to the camelback slings.
- The camelback slings must be modelled with 5m length and at a distance of 5m between each other. The slings standard positions are (14m, 19m) or (20m, 25m). These values can be changed if negotiated with the project.
- The static configuration overlength must be bigger than 1m and lower than 4m.

5.4.3.7 GENERAL CASES SUMMARY

The general cases are summarized in Table 2.

Table 2 - DVC Analysis Cases Summary

Phase	Load Case	Description
Installation	Case 1	ELT analytical analysis
	Case 2	Equilibrium condition, DVC static configuration
	Case 3	VCM fixed in the HUB:
		(i i) Suspended line with vertical motion of vessel (heave-up)
		(i i) Flexible lay, moment when flexible touches the seabed
Offshore Test	Case 4 With Buoys	Hydrostatic Test, 110% of project pressure. VCM connected to hub and flexible completely layed. Buoyancy modules (or deadweights) still attached to flexible.
	Case 4 Without Buoys	Hydrostatic test, 110% project pressure. VCM fixed at hub and flexible completely layed. Buoyancy modules (or deadweights) released.
Operation	Case 5	Operation condition, 100% operation or project pressure. VCM fixed to HUB and flexible completely layed. Buoyancy modules (or deadweights) released.

The Figure 17 shows a typical bending moment curve behaviour over time for all DVC quasi-static cases.

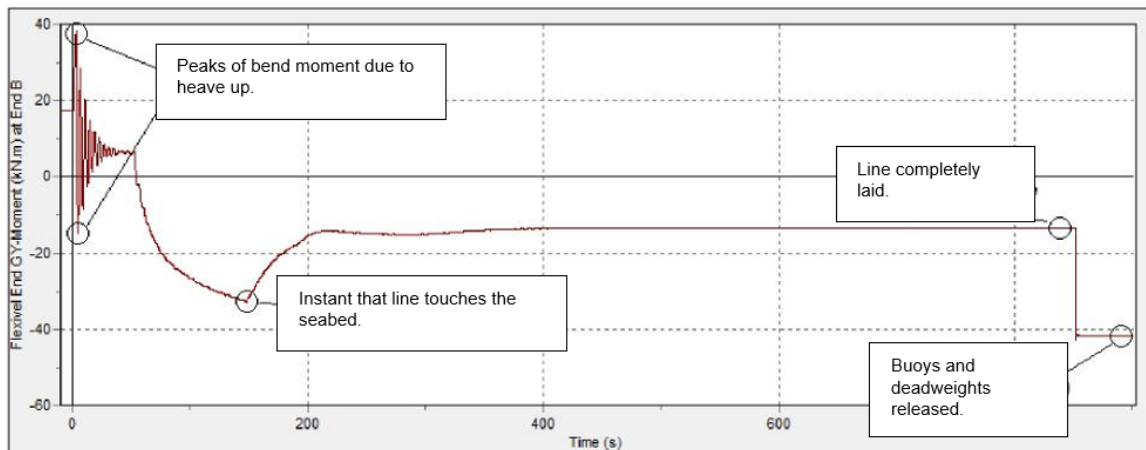


Figure 17: Typical Bending Moment Curve Over Time (Cases 2 to 5)

The competed flange results must be presented as **Appendix 3 – Static and quasi-static VCM Flange results presenting.**

5.4.4 CONTINGENCY ANALYSIS

5.4.4.1 FLEXIBLE LINE LIFT DURING THE SOFT LANDING

For performing contingencies assessment, an uplift must be applied in any point along the flexible line. The steps to perform the contingency cases are summarized below:

- With the static configuration found in DVC cases 2 and 3, the VCM must be fixed.
- Flexible line must be paid out until it touches the seabed. A best practice is to increase the flexible length by around 8m (other value can be adopted to ensure the flexible line is touching the seabed).
- Add up to 2000kg of buoys in any point of the flexible line (example: 2m, 3m or 4m from the flange). It is mandatory that there is no DVC buoy previously installed at this point. The minimum distance from the flange to apply contingency uplift is 1m.
- Run static analysis and check if any analysis criteria are being exceeded, as flexible line MBR and bend restrictor loads. If any limiting criteria is being infringed, reduce the uplift value or change distance from the flange.

This contingency can be used for 1st and 2nd end DVC. At least two options of contingency uplifts must be assessed.

5.4.4.2 MAXIMUM CAMELBACK PULL

For 2nd end DVC configuration, the camelback shall be used to check the maximum pull that the line can accept. The steps to perform this contingency are summarized below:

- The VCM must be kept free in static configuration found in cases 2 and 3.
- The A&R wire must be paid in (recovered). The VCM maximum angle to be reached is 6°.
- Run static analysis and check if any analysis criteria are being exceeded, as flexible line MBR and bend restrictor loads. If any limiting criteria is being infringed, reduce the camelback pull and consequently the VCM angle.

5.4.5 1ST END DVC DYNAMIC ANALYSIS

The DVC dynamic analysis is done in 3 steps as shown in Figure 18. The steps are:

5.4.5.1 VCM AT 50 M FROM THE SEABED

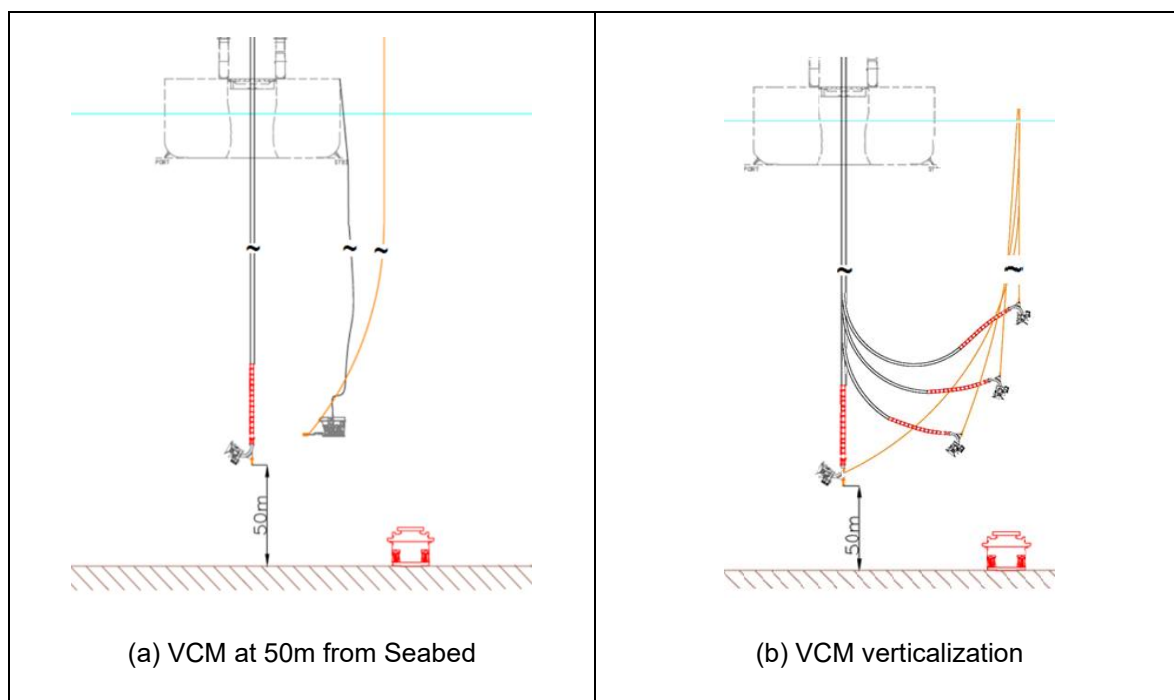
In this step VCM is hanging at 50m from the seabed with no buoyancy module installed as shown in Figure 18 (a).

5.4.5.2 VCM VERTICALIZATION (CRITICAL STEP)

In this step the VCM is verticalized by PLSV crane wire as shown in Figure 18 (b). The buoyancy modules must be installed, if required by the static configuration. A line feeding or a static step by step reducing the crane wire length should be done to find the critical step for the bend restrictor loads (bend moment and shear force). The worst steps must be considered for the dynamic analysis. The AHC can be considered turned on to respect the analysis criteria.

5.4.5.3 VCM CONNECTED TO THE SUBSEA EQUIPMENT

In this step VCM is already connected to the subsea equipment with the same configuration found in static analysis as shown in Figure 18 (c).



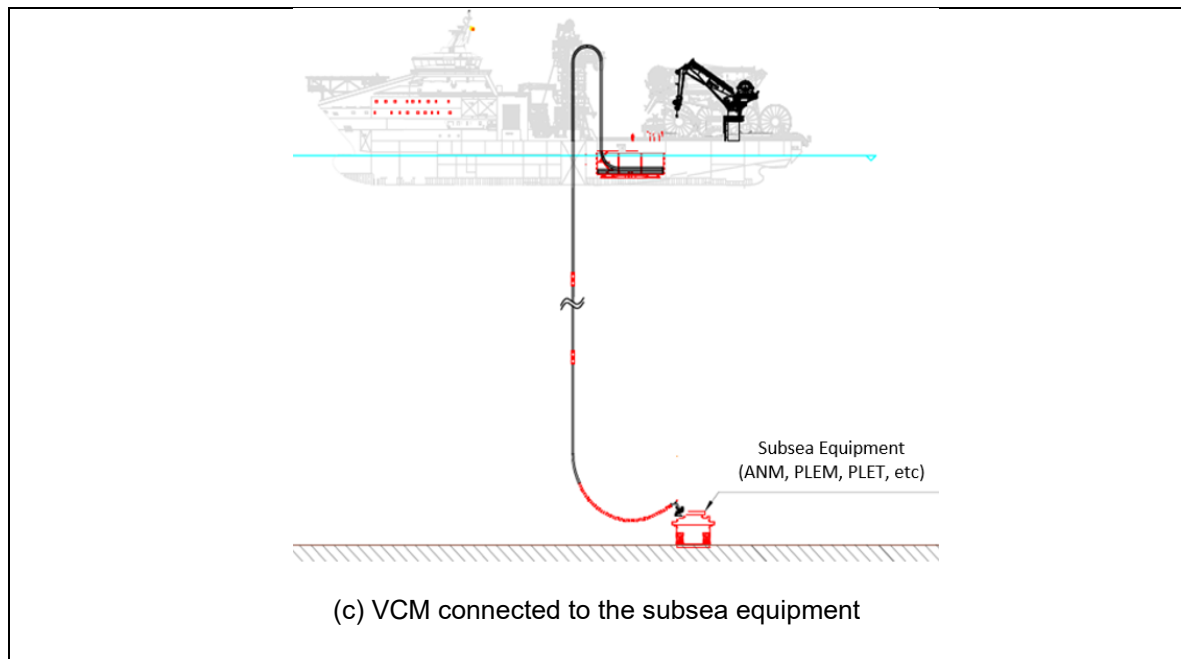


Figure 18: 1st end DVC Dynamic Steps

The post processing must be done for the latest wave period, to avoid numerical peaks.

5.4.6 2ND END DVC DYNAMIC ANALYSIS

For the 2nd end DVC, the static configuration must be done in advance to check whether is required to consider buoyancy modules (or dead weight) or not. Both dry and flooded annulus must be considered. The 2nd end DVC dynamic steps are shown in Figure 19. The steps are:

5.4.6.1 END FITTING AT WORKTABLE

In this step the flexible line is hanging at the worktable with no buoyancy modules waiting for the VCM flange connection as shown in Figure 19 (a). In Orcaflex, the end fitting is required to be modeled as fixed in the worktable. In this step, the PLSV is free to weathervane and Near/Far approach shall be considered.

5.4.6.2 VCM OVERBOARDING

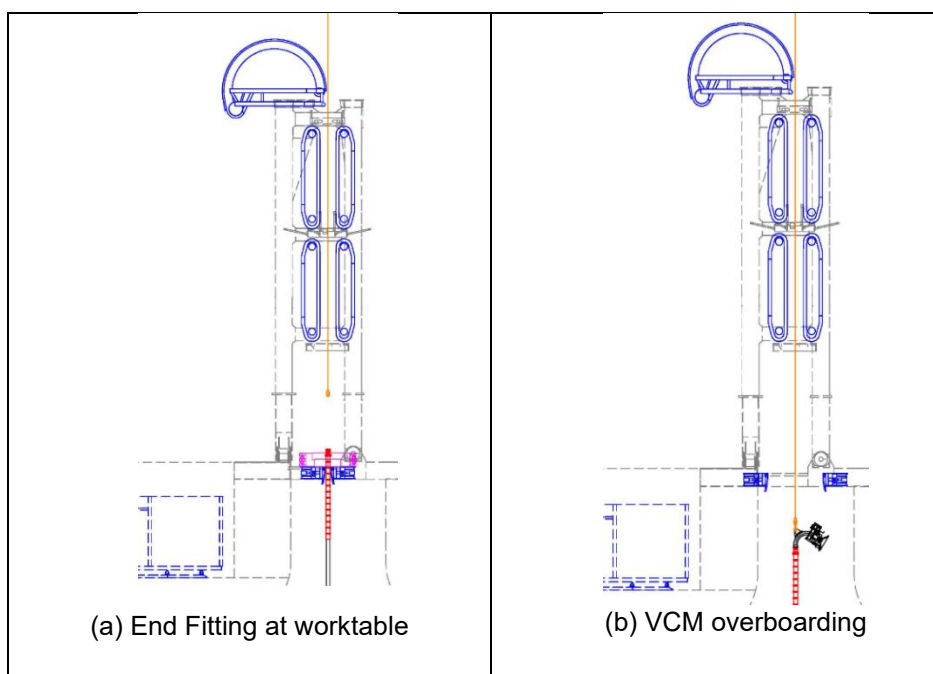
In this step VCM supports whole catenary weight, with no buoyancy modules (or dead weight) installed as shown in Figure 19 (b). In this step, the PLSV is free to weathervane and Near/Far approach shall be considered.

5.4.6.3 WAITING WINDOW

PLSV is waiting for installation window in the worst case scenario: with the A&R still connected to the VCM shackle (before the load transfer from A&R to crane wire), as shown in Figure 19 (c). For the static assessment, the bend radius at TDP shall be within the limit, and A&R top angle can be changed.

5.4.6.4 VCM CONNECTED TO THE SUBSEA EQUIPMENT

In this step VCM is already connected to the subsea equipment with the same configuration found in static analysis as shown in Figure 19 (d).



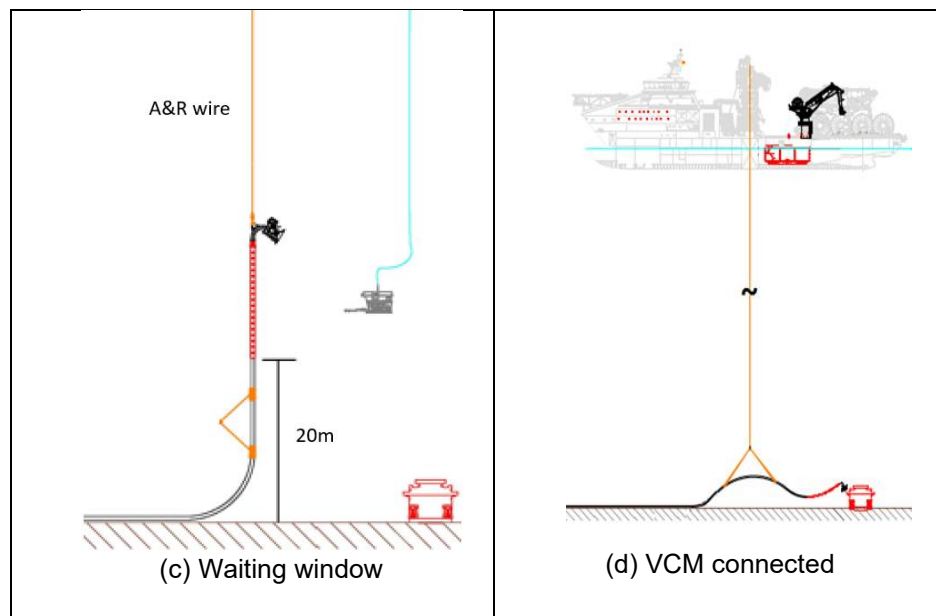


Figure 19: 2nd end DVC Dynamic Steps

If the project bend restrictor is long enough to cause concern at the camelback formation step, an extra step may be considered as shown in section 5.4.6.5. The post processing must be done for the latest wave period, to avoid numerical peaks.

5.4.6.5 BEND RESTRICTOR SETTLEMENT

In this step it is required to assess if the Bend radius is not beyond the limit. The step must be done setting the bend restrictor last node to the TDP as shown in Figure 20.

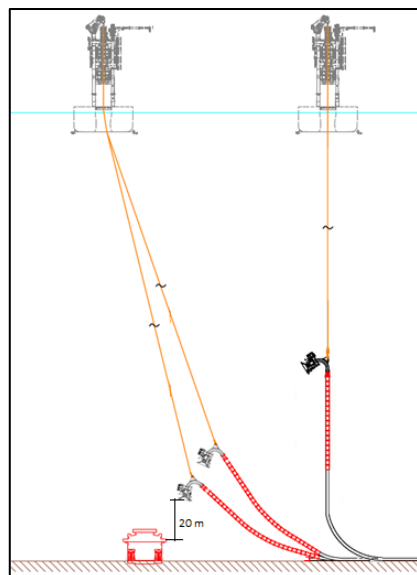


Figure 20: 2nd end DVC Extra Dynamic Step – Bend Restrictor Settlement

5.5 TOOLS

5.5.1 LTC TOOL - DVC SPREADSHEET

Following quality and standardization criteria, pre- and post-processing must be carried out using tools developed by the IAS department. The spreadsheet covers both DVC analyses: 1st and 2nd end.

Notes:

Note 1: Pay attention to the tool version used and keep the python libraries updated.

Note 2: These tools are commonly used for LTC projects analysis, covering just DVC flange nominal height (+/- 52cm not included).

Note 3 :Use the sheets “Values”, “MCV” e “Guindaste”, “Bending Stiffness and Script Table” for pre-processing.

Note 4: Use the sheets “Results”, “Esforços” and “Relatório” for post-processing and report elaboration.

Note 5: Not even all inputs will be necessary, depends on the project data.

5.5.2 SUPPLY AND EPCI TOOL - DVC AUTOMATE (UNDER DEVELOPMENT)

DVC Automate is commonly used for Supply and EPCI projects analysis. This tool works in a similar way to the DVC spreadsheet as seen in Section 5.5.1 but involving a wider range of dynamic cases. In this tool, its necessary input the structures data, VCM data, Stiffness Curves, Capacity Curves and Environment data.

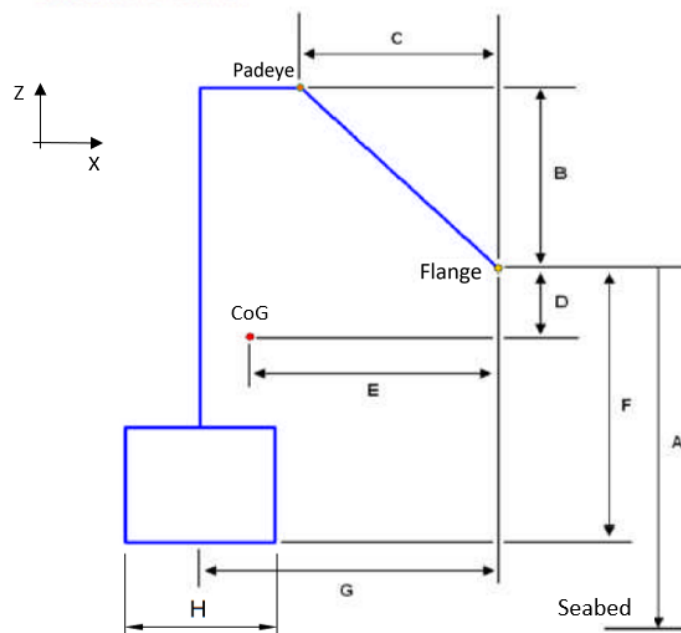
6. RECORDS

“Not Applicable”

7. APPENDICES & FORMS

7.1 APPENDIX 1 – VCM DATA

VCM data		Values
Mass		
Submerged apparent weight		
Swivel		
α	Gooseneck Angle	
A	Vertical Distance from flange to seabed	
B	Vertical Distance from padeye to flange	
C	Horizontal Distance from padeye to flange	
D	Vertical Distance from flange to CoG	
E	Horizontal Distance from flange to CoG	
F	Vertical Distance from flange to the base	
G	Horizontal Distance from flange to hub center	
H	Main Width	
I	CoG position in relation to Y axis	



7.2 APPENDIX 2 – BEND RESTRICTOR BENDING STIFFNESS CALCULATION.

As per Ref. [2], the bend stiffness curve of the bend restrictor is calculated as:

Table 3 - Bending Moment Curve for Bend Restrictor

Curvature	Bending moment
0	0
$\frac{1}{\text{Locking radius}}$	0,01
$1 + \frac{1}{\text{Locking radius}}$	0,01 + EI

The bending stiffness EI is calculated as:

$$EI = E * \frac{\pi * (D_{ext}^4 - D_{int}^4)}{64}$$

Where D_{ext} and D_{int} are external and Internal diameter of the bend restrictor smallest cross section, respectively, and E is the young modulus. For young modulus, standard values considered are:

- For metallic bend restrictors (steel): $E = 207 * 10^6 \text{ kN/m}^2$
- For PU bend restrictors: $E = 7 * 10^6 \text{ kN/m}^2$

7.3 APPENDIX 3 – STATIC AND QUASI-STATIC VCM FLANGE RESULTS PRESENTING.

For EPCI and Supply phases, the VCM flange complete results must be presented as Table 4. Note that, the loads are extracted based on the bend moment (maximum or minimum) and its equivalent axial and shear forces (values at the same simulation time). For cases 4 and 5, the loads must be in accordance with Equation 1. For case 3ii only the maximum bend moment is considered.

Table 4 - VCM Flange Loads Results Presenting.

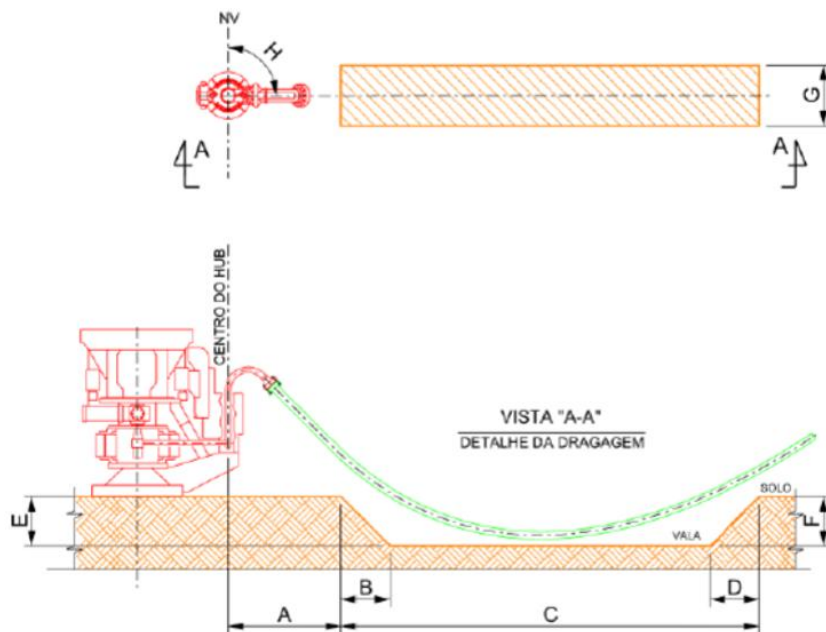
Case		Load	Value	Unit
ELT (Case 1)		Axial Force		kN
1 st end DVC - Equilibrium (Case 2)		Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
		MBR (Bend restrictor)		m
		MBR (Flexible)		m
1 st end DVC - Heave-up with VCM connected (Case 3i)	Minimum Bend Moment	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
	Maximum Bend Moment	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
	MBR (Bend restrictor)			m
	MBR (Flexible)			m
1 st end DVC – flexible lay away (Case 3ii)	(a)	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
	(b)	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
1 st end DVC - Test Offshore (Before buoys removal) (Case 4)	(a)	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
	(b)	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
1 st end DVC - Test Offshore (After buoys removal) (Case 4)	(a)	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
	(b)	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
1 st end DVC - Operation (Case 5)	(a)	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm
	(b)	Axial Force		kN
		Shear Force		kN
		Bending Moment		kNm

Notes:

(a) – Analysis considering Flange height +52cm.

(b) – Analysis considering Flange height -52cm.

7.4 APPENDIX 4 – DREDGING QUOTAS



DREDGING DATA		VALUES
A	DISTANCE FROM THE CENTER OF THE HUB TO THE TRENCH START	
B	DECLINE 1 HORIZONTAL DISTANCE	
C	TOTAL TRENCH LENGTH	
D	DECLINE 2 HORIZONTAL DISTANCE	
E	DEPTH 1	
F	DEPTH 2	
G	WIDTH (Not required for Orcaflex modelling)	

Notes:

Note 1: Mandatory to keep maximum depth of 1m (data E & F).

Note 2: Recommended at least 1m of distance from the center of the HUB to the trench start (data A).

7.5 FORMS

“Not Applicable”